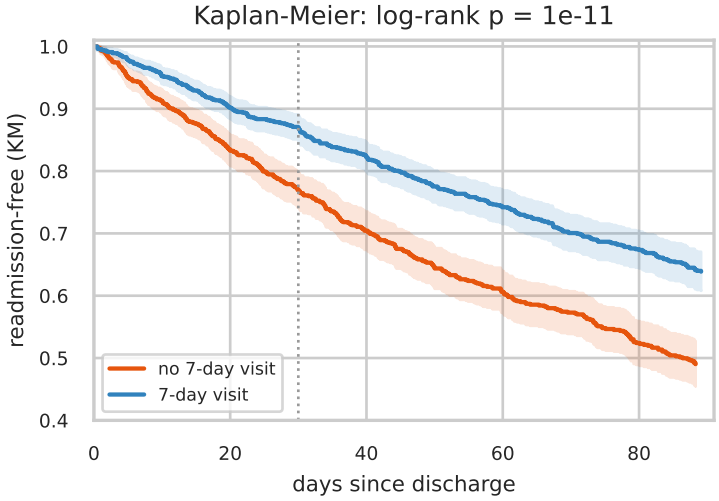
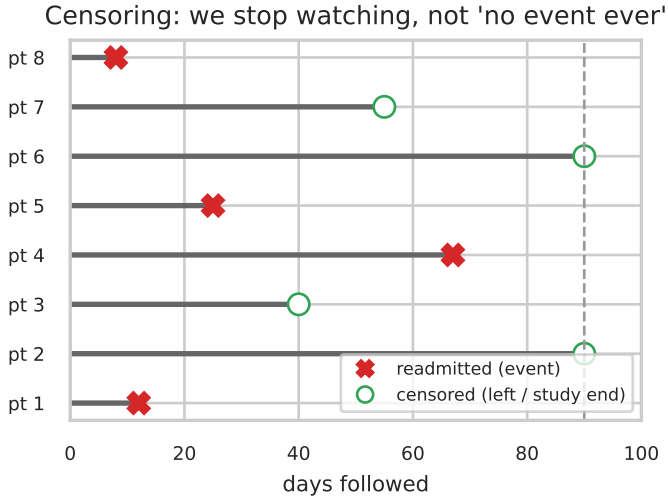
Day 13 Cheat Sheet: Survival Analysis

Time-to-event data: censoring, Kaplan-Meier, log-rank, Cox hazard ratios

TERM	WHAT IT MEANS	CODE (statsmodels)
Time-to-event	how long until something happens (readmit, death)	columns: days, readmitted (1/0)
Event	the outcome happened during follow-up	readmitted == 1
Censored	follow-up ended first; we only know "at least t"	readmitted == 0
Survival S(t)	share still event-free at time t	sm.SurvfuncRight(days, event)
Kaplan-Meier	S(t) = product of (1 - events / at risk)	sf.surv_prob, sf.surv_times
At risk	still followed and event-free just before t	sf.summary()["num at risk"]
Median time	t where S(t) drops to 0.5 (may not be reached)	sf.quantile(0.5)
Log-rank test	do two or more curves differ overall?	survdiff(days, event, group)
Hazard	instant event rate among those still at risk	(rate, not a probability)
Hazard ratio (HR)	hazard in group / reference; < 1 = lower risk	np.exp(cox.params)
Cox PH model	HRs adjusted for covariates	smf.phreg("days ~ x", df, status=e)
PH assumption	HR stays roughly constant over follow-up	fit by time window, compare HRs





THIS DATA (simulated, seed 13): 2,000 heart failure discharges, 90-day follow-up
Readmitted 761 (38.0%); censored 1239, of which 429 lost to follow-up before day 90
KM 90-day readmission: 42.8% (naive events/all says 38.0%)
30-day risk: no visit 23.1%, 7-day visit 13.1% | log-rank chi2 46.1, p = 1e-11
Cox HR: visit 0.61 (0.53-0.70), low EF 1.39, age 1.030 per year

HABITS THAT KEEP YOU HONEST
- Never average "days" across patients when some were censored; use KM
- Never drop censored patients; they are still at risk until they leave
- Report risk at a fixed time (30 or 90 days) along with the curve
- An HR is a ratio of rates, not of probabilities: say "lower hazard", not "39% fewer patients"
- Observational: a 7-day visit is linked to fewer readmissions; healthier patients may go more often

```
import statsmodels.api as sm, statsmodels.formula.api as smf
sf = sm.SurvfuncRight(df.days, df.readmitted)            # Kaplan-Meier
stat, p = sm.duration.survdiff(df.days, df.readmitted, df.followup_visit)
cox = smf.phreg("days ~ followup_visit + age + low_ef", df, status=df.readmitted).fit()
np.exp(cox.params)                                       # hazard ratios
```

Infographic: How Fast Do Heart Failure Patients Come Back?

Simulated: 2,000 discharges, 90-day follow-up, survival analysis

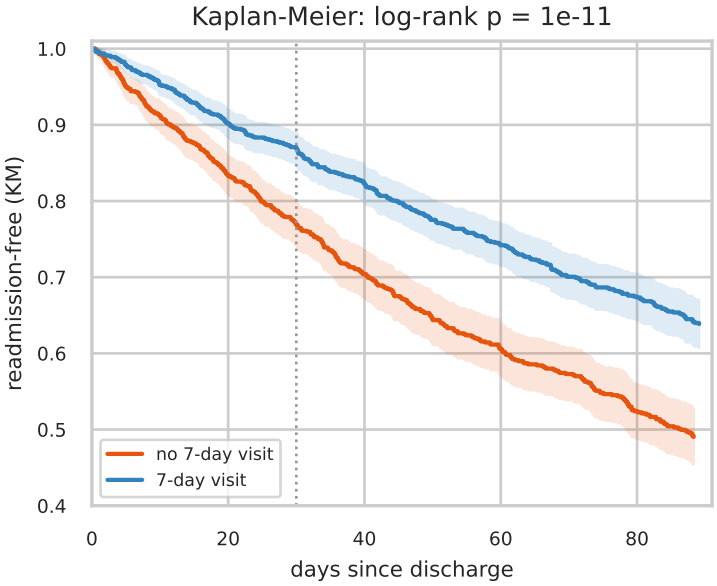
Define	Flag	KM curve	Log-rank	Cox	Report
start, event, end of watch	event 1 or censored 0	S(t) per group	curves differ overall?	adjusted hazard ratios	risk at 30d + the curve

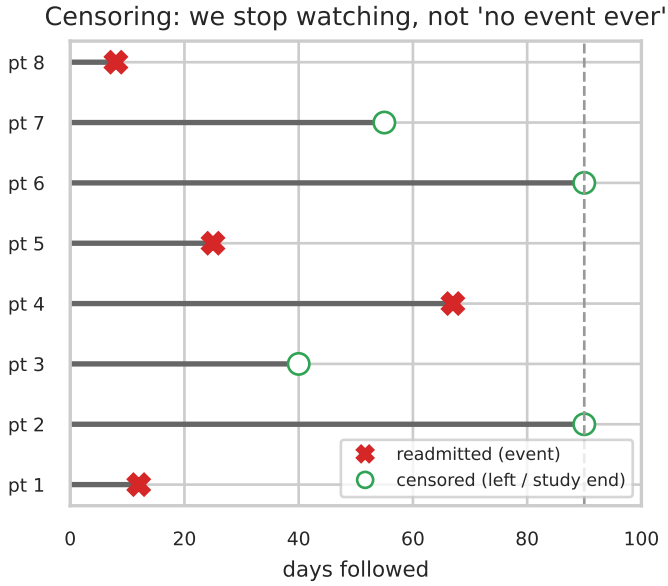
CENSORED
62% of patients not an event

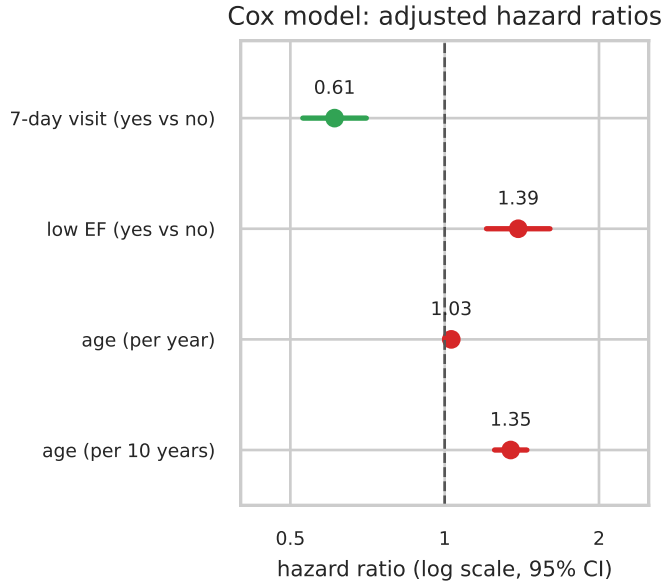
30-DAY RISK
23% vs 13% no visit vs visit

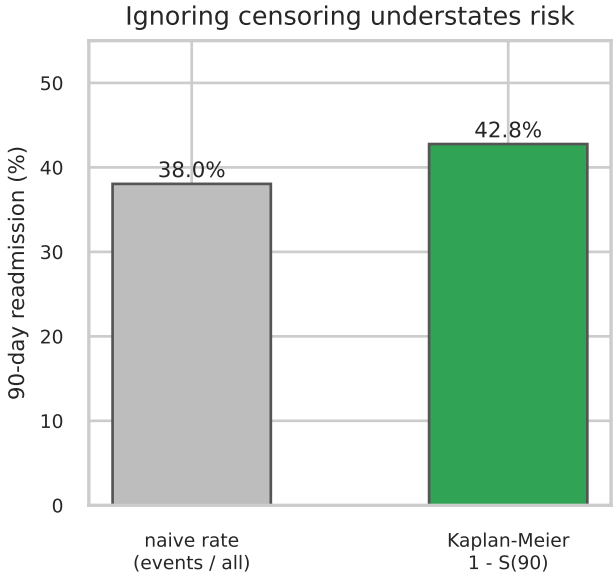
COX HR
0.61 7-day visit

LOG-RANK
p = 1e-11 curves differ









Read it like a care manager: patients seen within 7 days of discharge came back less often, and the gap opens early.
After adjusting for age and ejection fraction the readmission hazard is about 39% lower.
This is observational and simulated: it supports a follow-up program, it does not prove one works.

Hands-On Worksheet: ds-genai-90-days / day13

60-90 minutes | notebook: day13/survival_readmissions.ipynb

[ ] mkdir day13, conda activate ds90, open survival_readmissions.ipynb

[ ] Simulate hf_readmissions.csv with seed 13 (snippet 1)

[ ] Count events vs censored; how many left before day 90?

[ ] Compare naive mean days vs mean among readmitted (snippet 2)

[ ] Build Kaplan-Meier by hand: at_risk, events, cumprod

[ ] KM with statsmodels per group; read 30- and 90-day values (snippet 3)

[ ] Plot both step curves with a day-30 marker

[ ] Log-rank test for the 7-day visit (snippet 4)

[ ] Cox model: HRs for visit, age, low EF with 95% CIs

[ ] PH check: visit HR in days 0-30 vs 30-90 (snippet 5)

[ ] Translate to 30-day risk per group for a care manager

[ ] Findings cell: 5 bullets + limits (observational, simulated)

[ ] git add day13 && git commit && git push (snippet 6)

[ ] Stretch: Snowflake SQL life table by week (snippet 7)

```
import pandas as pd, numpy as np, statsmodels.api as sm
df = pd.read_csv("hf_readmissions.csv")
sf = sm.SurvfuncRight(df.days, df.readmitted)
print(1 - sf.surv_prob[sf.surv_times <= 30][-1])        # 30-day risk
cox = sm.PHReg.from_formula("days ~ followup_visit + age + low_ef", df, status=df.readmitted).fit()
print(np.exp(cox.params))
```

Events: _____ Censored: _____ Lost early: _____ KM 90-day risk: _____ Naive rate: _____

30-day risk no visit: _____ visit: _____ Log-rank p: _____ HR visit: _____ HR low EF: _____

Day 13 Self-Check Quiz

1. A patient moved away on day 40 with no readmission. Event, censored, or drop them?

_____

2. Why is the average of 'days' across all patients misleading here?

_____

3. In words, how does Kaplan-Meier update S(t) at each event time?

_____

4. The no-visit group's median is about 86 days. Why can't we give one for the visit group?

_____

5. What does the log-rank test tell you, and what does it not?

_____

6. HR = 0.61 for the 7-day visit. Say it in one plain sentence.

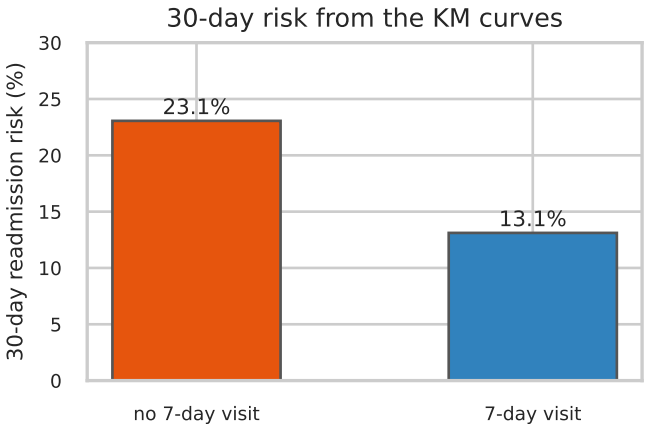
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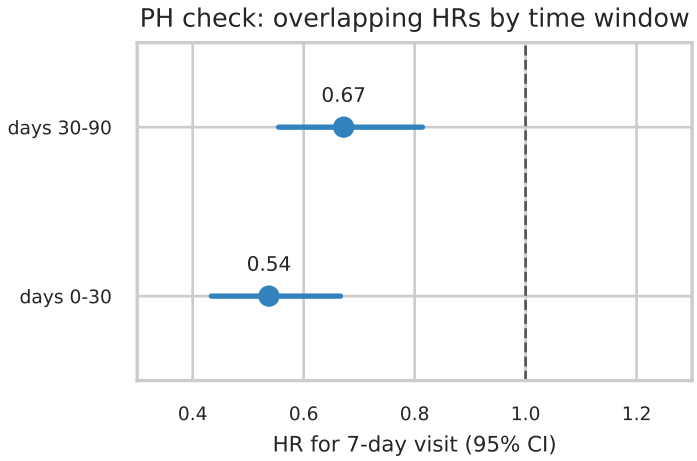
7. What does the proportional hazards assumption mean, and how did we check it?

_____

8. Does this prove 7-day visits cut readmissions? What else could explain it?

_____





Answers:
1 censored (readmitted = 0, days = 40): they count as at risk up to day 40, then leave the risk set. Dropping them biases the curve
2 censored patients' days are "at least t", not their true time; mixing them in (or dropping them) gives a number that means nothing
3 S(t) = S(previous) x (1 - events at t / patients still at risk just before t); censored patients shrink the risk set without a drop
4 the visit group's curve never falls to 0.5 within 90 days (64% still readmission-free), so its median is not reached
5 whether the curves differ overall (chi2 46.1, p = 1e-11); not how big the gap is, and it does not adjust for age or EF
6 at any point in follow-up, visit patients had about 39% lower readmission hazard than similar no-visit patients (CI 0.53-0.70)
7 the HR is roughly constant over time; we fit days 0-30 and 30-90 separately (0.54 vs 0.67, CIs overlap)
8 no: it's observational. Healthier or better-supported patients may be the ones who show up (confounding); a trial or careful adjustment is needed